ECE675

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Non-linear Model Predictive Traction Control

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1. Abstract

This report presents the design and implementation of a Nonlinear Model Predictive Controller (NMPC) for traction control of an electric vehicle. As modern electric vehicles (EVs) are capable of delivering high torque instantaneously, precise traction control is critical for both safety and efficiency. A planar 5-Degree-of-Freedom (5-DOF) dual-track vehicle model is developed, utilizing the Pacejka "Magic Formula" for nonlinear tire dynamics and combined slip conditions. The control problem is formulated as a reference-tracking NMPC, incorporating a secondary Nonlinear Program (NLP) to generate physically feasible steady-state reference trajectories. The controller was tested under adverse traction scenarios, including split-μ (friction) surfaces and sudden transitions to low-traction environments. While the NMPC successfully stabilized the vehicle and regulated wheel slip under these conditions, the solver performance indicates that the current nonlinear formulation is too computationally expensive for real-time application, suggesting the need for future linearization.

2. Introduction

2.1 Motivation

Vehicle traction control is a fundamental component of active safety systems. With the proliferation of high-torque electric powertrains, the risk of excessive wheel slip during acceleration is heightened. Proper management of this torque is required not only to maintain vehicle stability but also to maximize energy efficiency by operating tires within their optimal force-generation regions.

2.2 Goals

The primary goal of this project is to design a reference-tracking Nonlinear MPC capable of controlling the vehicle state under harsh traction conditions. Specifically, the controller aims to demonstrate stability and effective slip regulation in split-μ, low traction conditions where left and right tires experience drastically different friction coefficients.

3. System Modeling

3.1 Planar Vehicle Dynamics

The controller utilizes a dual-track model rather than a single-track "bicycle" model to account for lateral load transfer and independent rear-wheel drive dynamics. The vehicle frame has 3 degrees of freedom in the longitudinal (v_x), lateral (v_y) and yaw (r) directions. Longitudinal and lateral accelerations are caused by longitudinal and lateral forces generated at the four wheel-centers, and by aerodynamic drag which is negative and quadratic with forward velocity.

A net moment is generated by the wheel forces about the vehicle's center of mass which causes a yaw acceleration ($\frac{dr}{dt}$).

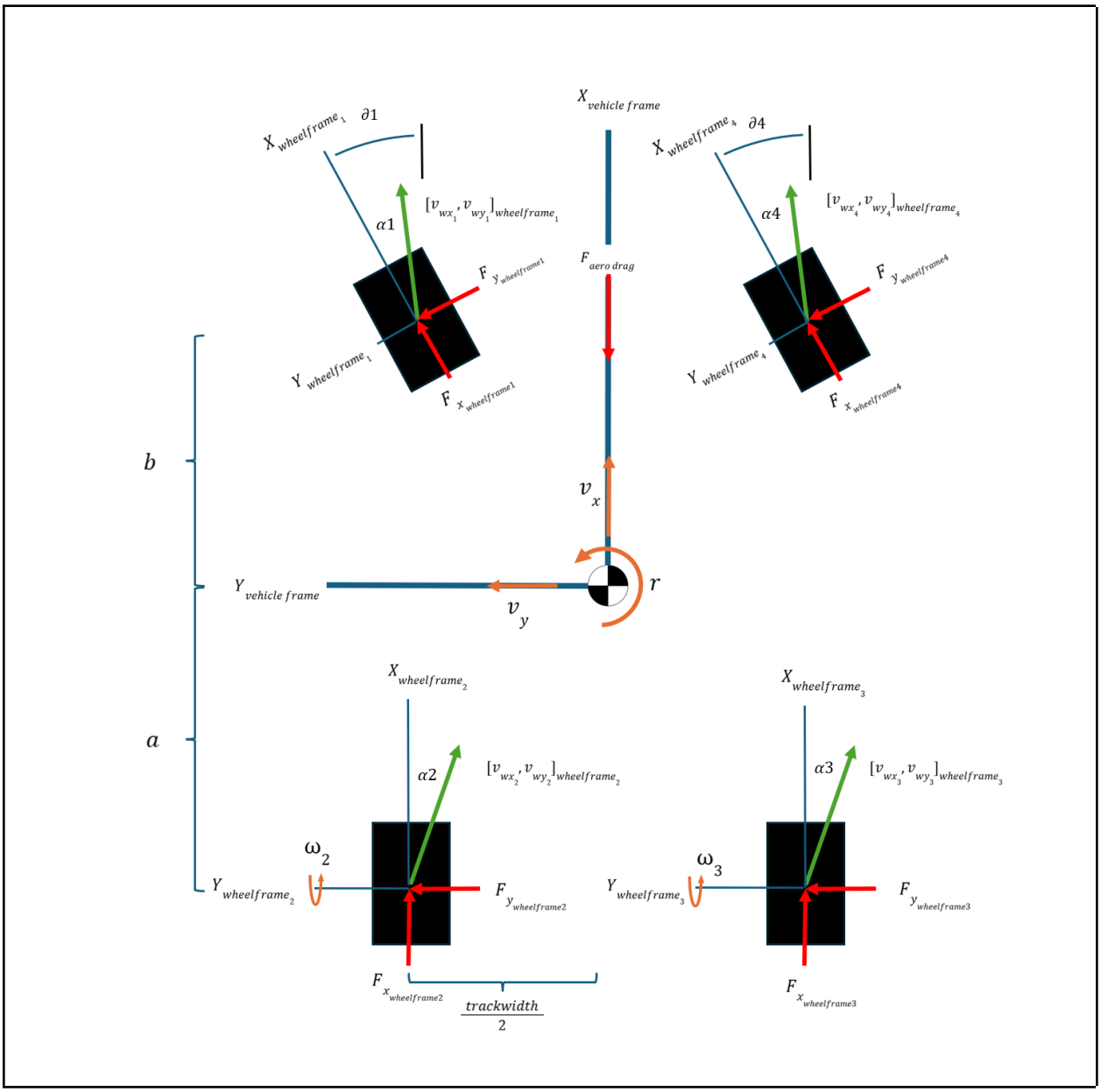

Figure 1: Dual-track model considers longitudinal and lateral forces at each tire. Longitudinal tractive force is a non-linear function of normal force.

$$\ddot{x} = \dot{y}r + \frac{F_{xfl} + F_{xfr} + F_{xrl} + F_{xrr} + F_{drag}}{m}$$

$$\ddot{y} = -\dot{x}r + \frac{F_{yfl} + F_{yfr} + F_{yrl} + F_{yrr}}{m}$$

$$\dot{r} = \frac{a(F_{yfl} + F_{yfr}) - b(F_{yrl} + F_{yrr}) + \frac{t_f}{2}(F_{xfl} - F_{xfr}) + \frac{t_r}{2}(F_{xrl} - F_{xrr})}{I_{zz}}$$

Figure 2: Differential equations for the planar vehicle's longitudinal, lateral and yaw dynamics

Steady-state load transfer is considered for both longitudinal (acceleration, braking) and lateral (cornering) dynamics. Load transfer for a given front-rear roll stiffness λ_{front} and vehicle center of gravity height h_{cg} are calculated in *Figure 3*.

Static corner loads

$$F_{z1,\text{stat}} = F_{z4,\text{stat}} = \frac{m\,g\,a}{2(a+b)}$$

$$F_{z2,\text{stat}} = F_{z3,\text{stat}} = \frac{m\,g\,b}{2(a+b)}$$

Longitudinal load transfer

$$\Delta F_{z,\text{long}} = m\,\dot{v}_x\,\frac{h_{cg}}{a+b}$$

Lateral load transfer

$$\Delta F_{z,\text{lat,front}} = \lambda_{\text{front}}\,m\,\dot{v}_y\,\frac{h_{cg}}{t}$$

$$\Delta F_{z,\text{lat,rear}} = (1-\lambda_{\text{front}})\,m\,\dot{v}_y\,\frac{h_{cg}}{t}$$

Final normal forces

$$F_{z1}\,(FL) = F_{z1,\text{stat}} - \Delta F_{z,\text{long}} - \Delta F_{z,\text{lat,front}}$$

$$F_{z2}\,(RL) = F_{z2,\text{stat}} + \Delta F_{z,\text{long}} - \Delta F_{z,\text{lat,rear}}$$

$$F_{z3}\,(RR) = F_{z2,\text{stat}} + \Delta F_{z,\text{long}} + \Delta F_{z,\text{lat,rear}}$$

$$F_{z4}\,(FR) = F_{z1,\text{stat}} - \Delta F_{z,\text{long}} + \Delta F_{z,\text{lat,front}}$$

Figure 3: Steady-state load transfer

3.2 Drivetrain & Tire Mechanics

Rear tire dynamics (ω_2, ω_3) add 2 additional degrees of freedom to the model (*Figure 4*). Front tires are assumed to have no longitudinal slip, so their angular speed can be determined algebraically from the vehicle velocities (*Figure 5*).

$$I_2 * \acute{\omega}_2 = T_{motor,left} - B_2 * \omega_2 - T_{load,left}$$

$$I_3 * \acute{\omega}_3 = T_{motor,right} - B_3 * \omega_3 - T_{load,right}$$

$where$:

$$T_{load,left} = r_{wheel} * F_{yrl}$$

$$T_{load,right} = r_{wheel} * F_{xrr}$$

Figure 4: First order rear wheel drivetrain dynamics for left rear (ω_2) and right rear (ω_3) wheels

$$\omega_1 = ((v_x - \frac{trackwidth}{2} * r) * cos(\delta) + (v_y + b * r) * sin(\delta))/r_{wheel}$$

$$\omega_4 = ((v_x + \frac{trackwidth}{2} * r) * cos(\delta) + (v_y + b * r) * sin(\delta))/r_{wheel}$$

Figure 5: Front left (ω_1) and front right (ω_4) wheel speeds determined from vehicle frame velocity

Longitudinal slip (λ) and lateral slip (α) at any instant can be determined from known vehicle velocities (v_x, v_y, r) and angular wheel velocities (ω_i). Longitudinal slip λ is found by comparing the tire radius speed($w_i * r_{wheel}$) to the linear velocity of the wheel in the rolling direction ($v_{wx,i}$), where $\lambda > 0$ tends toward positive vehicle acceleration. Lateral slip α is the angle between the wheel's rolling direction (δ for front wheels) and the wheel center velocity vector $\left[v_{wx,i}, v_{wy,i}\right]_{wheelframe_i}^{T}$. See *Figure 6*.

Longitudinal slip ratio (all 4 wheels)

$$\lambda_i = \frac{\omega_i r_w - v_{wx,i}}{v_{wx,i} + \varepsilon} \qquad i = 1,2,3,4$$

Lateral slip angle (all 4 wheels)

$$\alpha_i = \text{atan}\left(\frac{v_{wy,i}}{v_{wx,i}}\right) \qquad i = 1,2,3,4$$

where

$v_{wx,i}$: wheel i longitudinal velocity in wheel frame
$v_{wy,i}$: wheel i lateral velocity in wheel frame
r_w : effective tyre radius
$\varepsilon = 10^{-6}$: small constant to prevent division by zero

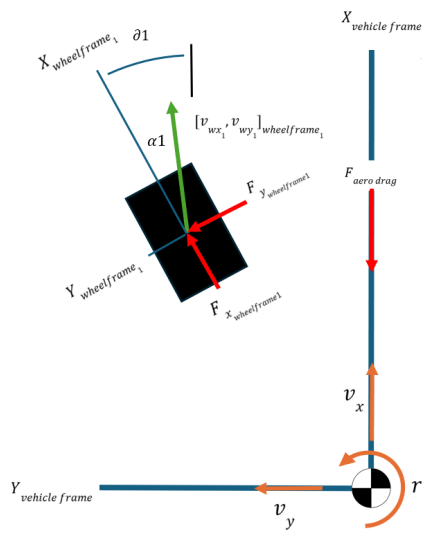

Figure 6: Vehicle model state and input vector

Pacejka "magic formula" was used for finding wheel frame forces from the known slip ratios λ_i, α_i. This formula is semi-empirical and highly nonlinear. Realistic values have been chosen for this model and subsequent controller design, but for implementation in a real vehicle, tire test data should be used to tune the shaping parameters B, C, E for best fit in both slip directions. See the empirical formulas in *Figure 7*, and the slip-force plots for the chosen vehicle parameters in *Figure 8*. Vehicle parameters are listed in *Appendix*.

$$F_{long_{0i}} = \mu_{long_i} * F_{z_i} * sin(C_{long_i} * atan(B_{long_i} * \lambda_i - E_{long_i} * (B_{long_i} * \lambda_i - atan(B_{long_i} * \lambda_i))))$$

$$F_{lat_{0i}} = \mu_{lat_i} * F_{z_i} * sin(C_{lat_i} * atan(B_{lat_i} * \alpha_i - E_{lat_i} * (B_{lat_i} * \alpha_i - atan(B_{lat_i} * \alpha_i))))$$

$$F_{long_i} = \frac{F_{long_{0i}}}{\varepsilon + \sqrt{1 + (F_{lat_{0i}} / \mu_{lat_i})^2}}$$

$$F_{lat_i} = \frac{F_{lat_{0i}}}{\varepsilon + \sqrt{1 + (F_{long_{0i}} / \mu_{long_{0i}})^2}}$$

Figure 7: Pacejka Magic Formula, where F_{long_i}, F_{lat_i} are wheel frame forces

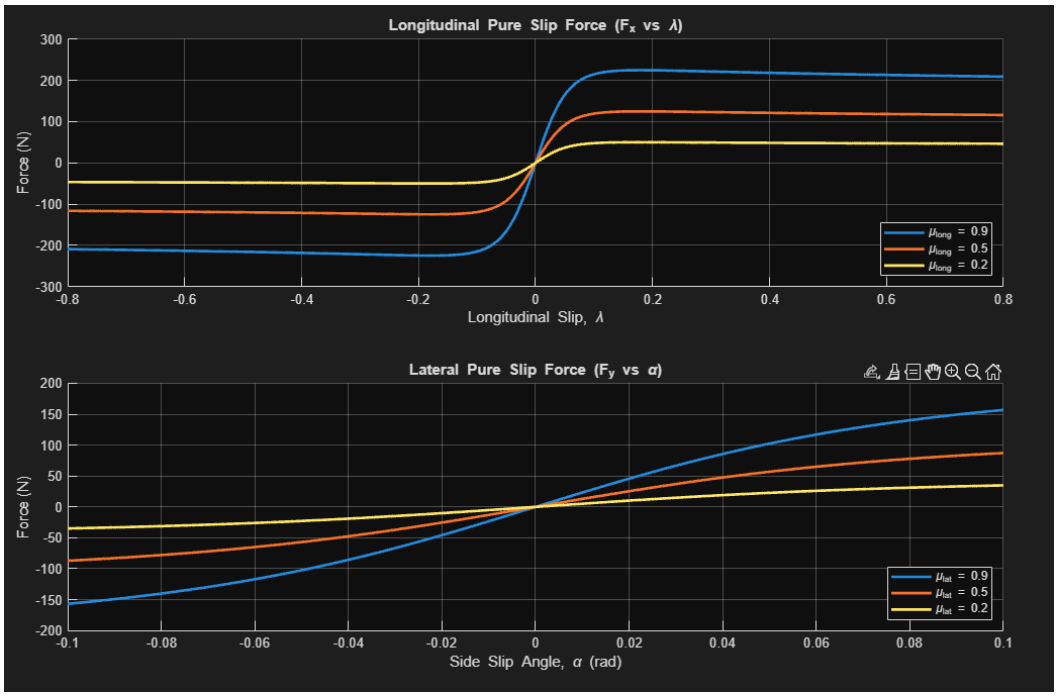

Figure 8: Magic formula (slip vs force) plotted for chosen vehicle parameter

3.3 Full Vehicle Model

It follows that the full model is 5th order, and represented by the state vector in *Figure 9*. Observability and observer design is not addressed in this paper, and it is assumed that accurate vehicle state values are readily available. Inputs to the model include front tire steer angle (δ) and left/right rear-wheel drive torques (T_{m2}, T_{m3}).

State vector

$$\mathbf{x} = \begin{bmatrix} v_x \\ v_y \\ r \\ \omega_2 \\ \omega_3 \end{bmatrix} \quad \begin{array}{l} \text{longitudinal velocity} \\ \text{lateral velocity} \\ \text{yaw rate} \\ \text{rear-left wheel speed} \\ \text{rear-right wheel speed} \end{array}$$

Input vector

$$\mathbf{u} = \begin{bmatrix} T_{m2} \\ T_{m3} \\ \delta \end{bmatrix} \quad \begin{array}{l} \text{torque on rear-left wheel} \\ \text{torque on rear-right wheel} \\ \text{front steering angle} \end{array}$$

Figure 9: Vehicle model state and input vector

3.4 Discrete DAE Solution

Due to the complexity of the combined-slip tire model, the system could not be explicitly solved for the derivative of the state vector (ODE formulation). Consequently, the model was converted to residual form and solved as a Differential Algebraic Equation (DAE, see *Figure 10*). The CasADi framework was used as an interface to solve these nonlinear DAEs using the IDAS integrator. The DAE solution is the next discrete state of the system x_{k+1} after the integration period T_s given an initial state x_k and input u_k.

$$\begin{aligned} \frac{d}{dt}v_x &= f_1(\frac{d}{dt}x,\ x,\ u) \\ \frac{d}{dt}v_y &= f_2(\frac{d}{dt}x,\ x,\ u) \\ \frac{d}{dt}r &= f_3(\frac{d}{dt}x,\ x,\ u) \\ \frac{d}{dt}\omega_2 &= f_4(\frac{d}{dt}x,\ x,\ u) \\ \frac{d}{dt}\omega_3 &= f_5(\frac{d}{dt}x,\ x,\ u) \end{aligned} \qquad 0 = \frac{d}{dt}x - f(\frac{d}{dt}x,\ x,\ u)$$

Figure 10: Implicit differential equations *(left)* and residual form for IDAS integration *(right)*

4. Controller Design

4.1 NMPC Formulation

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A nonlinear model predictive controller was formulated for reference tracking with quadratic cost. The optimization vector X_{opt} consists of 10 elements per horizon step N_h, which includes 3 inputs (steer angle δ, motor torques T_{m2}, T_{m3}), 5 states (velocities v_x, v_y, r and tire speeds ω_2, ω_3), and 2 slack variables *(Figure 11)*.

$$min_{X_{opt}} \sum_{i=1}^{N_h} [(x_i - x_{ref_i})^T * Q * (x_i - x_{ref_i}) + (u_i - u_{ref_i})^T * R * (u_i - u_{ref_i}) + S_{maxSlip_i}{}^2 * Q_{maxSlack} + S_{smallSlip_i}{}^2 * Q_{smallSlack}] + (x_i - x_{ref_i})^T * P * (x_i - x_{ref_i})$$

$$X_{opt} = [u_i, x_i, S_{maxSlip_i}, S_{smallSlip_i}]_{i=1}^{N_h}$$

$$s.t.$$

$$0 \le x_{i+1} - xNext(x_i, u_i) \le 0$$

$$x_{min} \le x_i \le x_{max}$$

$$u_{min} \le u_i \le u_{max}$$

$$-T_{\delta max} \le T_{2_i} - T_{3_i} \le -T_{\delta max}$$

$$-\lambda_{max} \le \lambda_1 - S_{maxSlip} \le \lambda_{max}$$

$$-\lambda_{max} \le \lambda_2 - S_{maxSlip} \le \lambda_{max}$$

$$-\lambda_{soft} \le \lambda_1 - S_{smallSlip} \le \lambda_{soft}$$

$$-\lambda_{soft} \le \lambda_2 - S_{smallSlip} \le \lambda_{soft}$$

Figure 11: Reference-tracking MPC formulation

Model dynamics are enforced over the horizon with the dual-inequality constraint in *Figure 11*, where $xNext()$ returns the discrete solution to the nonlinear DAE developed in section 3. Inequality constraints on x and u enforce limits on input torque, steering angle, and tire/vehicle velocities over the horizon. Slack variable $S_{smallSlip}$ is assigned a low cost $Q_{smallSlip}$ for exceeding 10% longitudinal slip. In this way, the controller attempts to drive the system to the reference state x_{ref} while using as close to the reference input u_{ref} as possible, but will depart from tracking the references when tire slip becomes excessive. A secondary slack variable $S_{maxSlip}$ with larger cost $Q_{maxSlip}$ applies a large penalty when wheel slip exceeds 20%. The controller tends to heavily cut wheel torque when slip activates the $S_{maxSlip}$ slack variable.

4.2 Reference Feasibility and Tracking

A critical challenge in reference tracking MPC is that the "desired" state is often physically impossible. For example, a user may desire a forward velocity $v_{x,\,desired} = 40m/s$ and no tire slip $\lambda_{i,\,desired} = 0$. However, to generate the longitudinal force required to maintain

$v_{x,\,desired}$ against air drag and drivetrain damping, a non-zero slip is physically required as $F_{long} > 0$ implies $\lambda > 0$ (see *Figure 7, 8*). If the provided reference x_{ref} is infeasible, the cost function will fight the dynamics. To solve this, a secondary Nonlinear Program (NLP) is run prior to the MPC solve. This NLP finds a feasible reference pair (x_{ref}, u_{ref}) such that:

$$x_{feasible} = xNext(x_{feasible}, u_{feasible})$$

Minimizing the error between $x_{feasible}$ and x_{ref} provided by the user. In this example with $v_{x,\,desired} = 40m/s$ and $\lambda_{i,\,desired} = 0$, the NLP calculates a feasible steady-state of $v_{x,\,feasible} = 39.7m/s$, $\lambda_{i,\,feasible} \approx 1\%$ and $97.6Nm$ of torque.

4.3 Cost Tuning

Costs were chosen to be diagonal and positive-definite. This includes the cost matrices Q (deviation from reference state x_{ref}), R (deviation from reference input u_{ref}), P (terminal deviation from reference state x_{ref}), $Q_{smallSlip}$(slack cost for $\lambda > 10\%$), and $Q_{maxSlip}$(slack cost for $\lambda > 20\%$). Starting weights were chosen based on the heuristic:

$$Q_i \approx \frac{100}{(Max\,\Delta x_i)^2} \qquad R_i \approx \frac{100}{(Max\,\Delta u_i)^2}$$

Which gives the initial values in *Table 1* below.

Index	Variable	Max Error (Δ Max)	$(\Delta\text{Max})^2$	Target Weight	**Proposed Weight**
Q1	V_x	30	900	$\frac{100}{900} \approx 0.11$	**0.1**
Q2	V_y	0.5	0.25	$\frac{100}{0.25} = 400$	**400**
Q3	Yaw Rate (r)	0.01	1e-4	$\frac{100}{1e\text{-}4} = 1,000,000$	$\mathbf{1,000,000}$
Q4	ω_2	$3 \times 30 = 90$	8100	$\frac{100}{8100} \approx 0.012$	**0.012**
Q5	ω_3	$3 \times 30 = 90$	8100	$\frac{100}{8100} \approx 0.012$	**0.012**
R1	T_{m2}	80	6400	$\frac{100}{6400} \approx 0.0156$	**0.016**
R2	T_{m3}	80	6400	$\frac{100}{6400} \approx 0.0156$	**0.016**
R3	δ	0.02	4e-4	$\frac{100}{4e\text{-}4} = 250,000$	$\mathbf{250,000}$

Table 1: Initial controller costs

Figure 12 below shows a $12ms$ simulation with *Table 1* controller cost matrices and no slack variables, from an initial forward velocity $v_x = 28m/s$ and reference forward velocity $v_{ref} \approx 40m/s$. The controller applies the reference torques to the left and right rear wheels, despite a longitudinal velocity error of $\sim 12m/s$. Cost Q_1 is far too low relative to R_1 and R_2, causing the controller to apply only the reference torque which is very slow to drive the vehicle states toward reference.

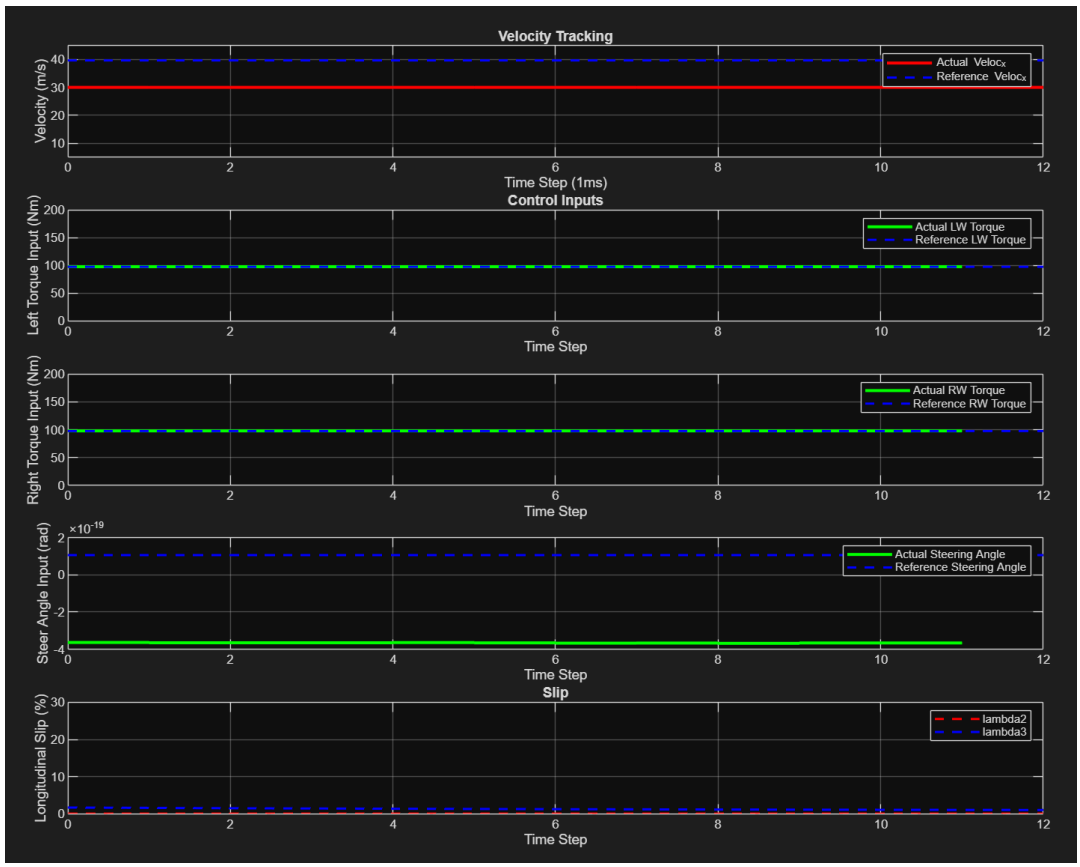

Figure 12: Initial simulation with initial $v_x = 28m/s$ and $v_{ref} \approx 40m/s$

Figure 13 shows a $375ms$ simulation with a much larger state-error cost ($Q_1 = 25e3$) and added slack costs $Q_{smallSlip} = 4e3$ and $Q_{maxSlip} = 10e3$ when 10% and 20% tire slips are exceeded. The controller saturates the torque input limit for both rear tires to $200Nm$ (see *Appendix* for all input constraint values). Small-slip slack costs become active as tire slip exceeds 10% and grow until input torque is reduced to avoid excessive wheel slip. The controller continues to drive toward x_{ref} with $160Nm$ of torque at each wheel and $\sim 16\%$ longitudinal tire slip.

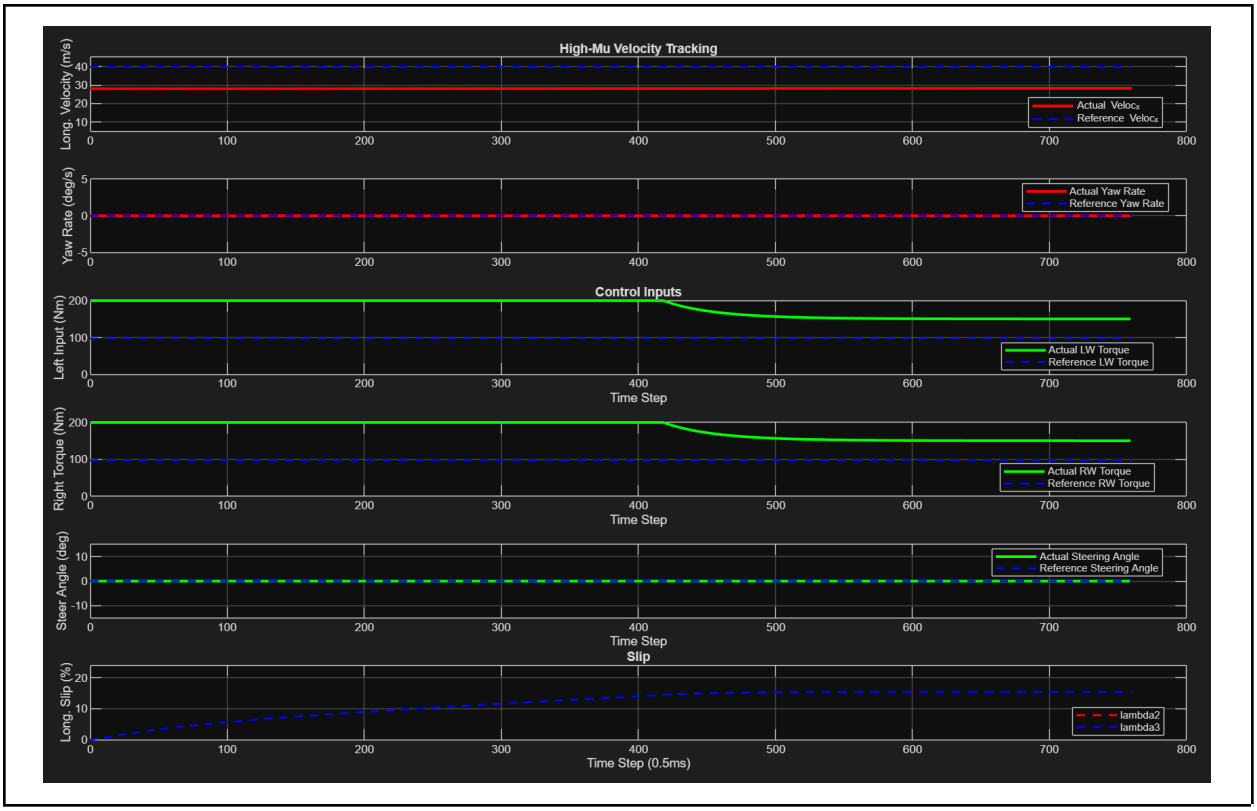

Figure 13: Simulation with initial $v_x = 28m/s$ and $v_{ref} \approx 40m/s$, and improved state-costs

Figure 14 below shows controller behavior with only a single slack cost which becomes active when the longitudinal slip exceeds 20%. The vehicle hits an ice patch ($\mu = 0.2$) on the left tires only while accelerating from an initial $v_x = 28m/s$ toward the reference velocity $v_{ref} \approx 40m/s$. When the 20% slip soft constraint is reached on the left wheel, left wheel torque is reduced heavily to manage wheel slip (~time step 425). The penalty for using non-reference input torques (R_1, R_2) were reduced for the subsequent simulations, which allowed the input torque to be more freely modulated by the MPC when encountering slippery conditions.

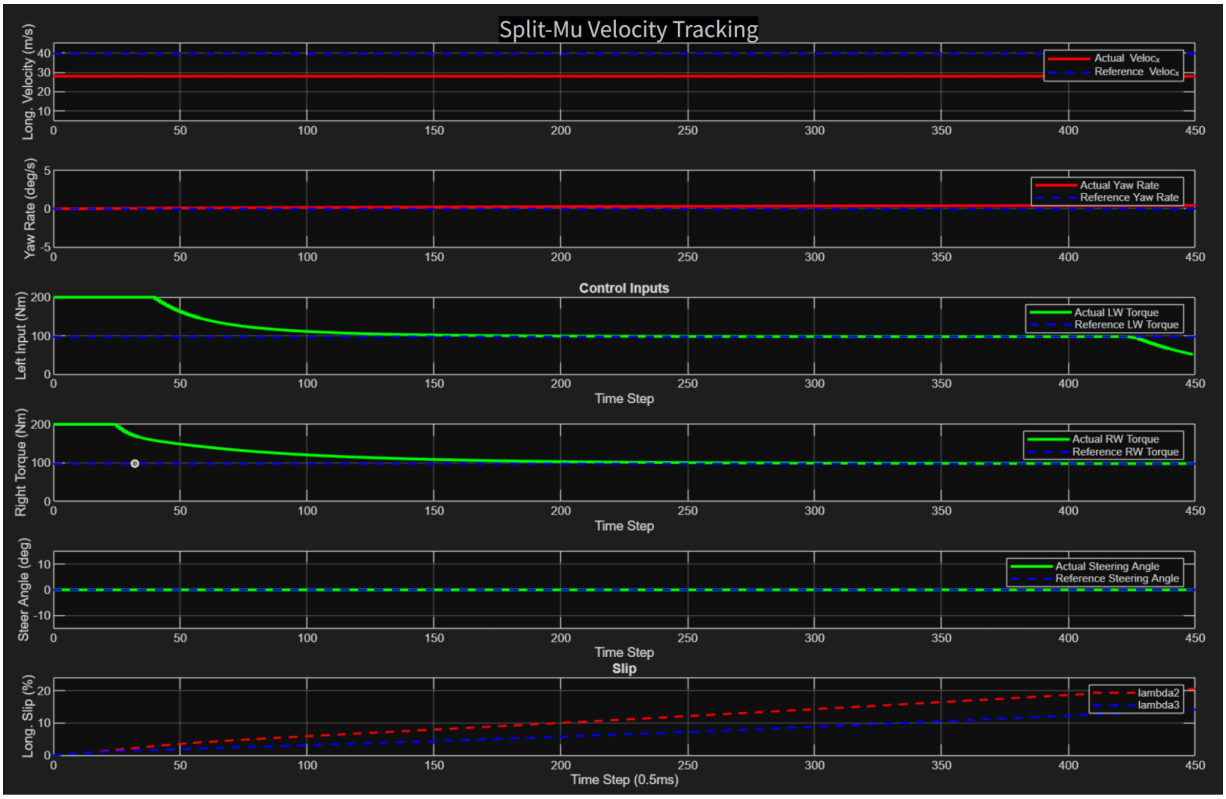

Figure 14: Accelerating from $v_x = 28m/s$ toward $v_{ref} \approx 40m/s$ with low left side friction ($\mu = 0.2$)

The reduced penalties R_1, R_2 allow the controller to more gradually reduce drive torques under the left-side low traction condition (*Figure 15*). Similar to the previous simulation (*Figure 14*), left and right wheel torque are both heavily reduced as the slip reaches 20%. For each simulation, the yaw rate is kept well under $0.5deg/s$

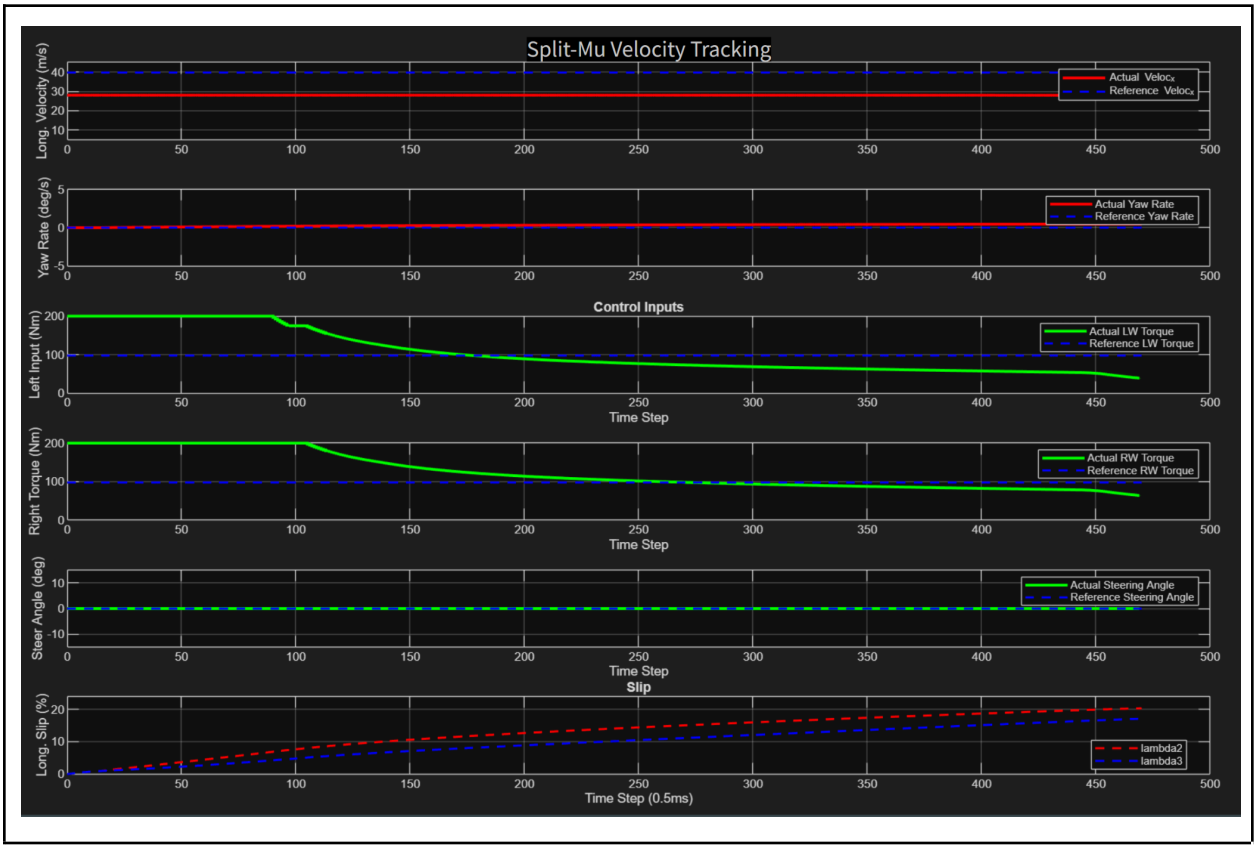

Figure 15: Accelerating from $v_x = 28m/s$ toward $v_{ref} \approx 40m/s$ with low left side friction ($\mu = 0.2$)

5. Conclusion

This study successfully demonstrated a Nonlinear Model Predictive Controller for traction control on a dual-track vehicle model. Through iterative tuning, it was established that a strategy combining high state-error penalties with reduced input costs allowed the controller to aggressively modulate torque, while a dual-tier slack variable formulation effectively managed slip limits.

Simulation results under adverse split-friction conditions confirmed the controller's ability to regulate independent wheel slip and maintain negligible yaw error (<$0.5deg/s$). However, the computational expense of the nonlinear DAE solution currently limits real-time viability. Future work should focus on deriving a linearized model for a Time-Varying Linear MPC or utilizing a Zero-Order Hold (ZOH) approach with reduced control frequency to achieve real-time performance, using this nonlinear framework as a high-fidelity benchmark.

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  title   = {{CasADi} -- {A} software framework for nonlinear optimization and optimal control},
  journal = {Mathematical Programming Computation},
  volume = {11},
  number = {1},
  pages   = {1--36},
  year   = {2019},
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Appendix

Vehicle Parameters:

Parameter Category	Symbol	Description	Value	Unit
Geometry	a	Distance from CG to Rear Axle	1.5	m
	b	Distance from CG to Front Axle	1.2	m
	t	Track Width	1.6	m
	h	Center of Gravity Height	0.55	m
	r_wheel	Wheel Radius	0.32	m
Inertial & Dynamics	m	Vehicle Mass	1000	kg
	Izz	Yaw Moment of Inertia	2400	kg*m^2
	Iw	Wheel Inertia (per wheel)	1.2	kg*m^2
	Bw	Wheel Damping Coeff.	0.01	Nms/rad
	λ_front	Roll Stiffness Distribution (Front)	0.45	-
Aerodynamics	ρ_air	Air Density	1.225	kg/m^3
	Cd	Drag Coefficient	0.28	-
	A_front	Frontal Area	2.2	m^2
	A_side	Side Area	2.5	m^2
Tire Model (Longitudinal)	Blong	Stiffness Factor	10	-
	Clong	Shape Factor	1.9	-
	Elong	Curvature Factor	0.97	-
	μ_long	Friction Coefficient	0.9	-
Tire Model (Lateral)	Blat	Stiffness Factor	8	-
	Clat	Shape Factor	1.3	-
	Elat	Curvature Factor	1	-
	μlat	Friction Coefficient	0.9	-
Controller Limits	Tmax	Max Torque Input	200	Nm
	δmax	Max Steering Angle	25	deg
	vx,max	Max Longitudinal Velocity	50	m/s

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Final Costs:

State Variable	Symbol	Stage Cost (Qii)	Terminal Cost (Pii)
Longitudinal Velocity (vx)	Q1,P1	25,000	0.2
Lateral Velocity (vy)	Q2,P2	400	800
Yaw Rate (r)	Q3,P3	1,000,000	2,000,000
Left Rear Wheel Speed (ω2)	Q4,P4	0.012	0.024
Right Rear Wheel Speed (ω3)	Q5,P5	0.012	0.024

Input Variable	Symbol	Value (Rii)
Left Rear Motor Torque (Tm2)	R1	0.0007
Right Rear Motor Torque (Tm3)	R2	0.0007
Steering Angle (δ)	R3	250,000